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SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> alter session set container=freepdb1;

```
SQL> alter session set container=freepdb1;

Session altered.

SQL>
```

Berpindah ke PDB bernama FREEPDB1

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Mengecek kembali dan mengonfirmasi bahwa sekarang berada di FREEPDB1

SQL> select username, account_status, common

from dba_users;

```
SQL> select username, account_status, common
from dba_users;
  2
USERNAME
--------------------------------------------------------------------------------
ACCOUNT_STATUS                   COM
-------------------------------- ---
SYS
OPEN                             YES

SYSTEM
OPEN                             YES

PDBADMIN
OPEN                             NO


USERNAME
--------------------------------------------------------------------------------
ACCOUNT_STATUS                   COM
-------------------------------- ---
SYSRAC
OPEN                             YES
```

```
DIP
LOCKED                         YES

DGPDB_INT
LOCKED                         YES


USERNAME
--------------------------------------------------------------------------------
ACCOUNT_STATUS                 COM
------------------------------ ---
SYSKM
LOCKED                         YES

SYS$UMF
LOCKED                         YES

SYSDG
LOCKED                         YES


36 rows selected.

SQL>
```

Menampilkan daftar semua user dalam FREEPDB1, status akun mereka, dan apakah mereka adalah user Common (YES) atau Local (NO)

SQL> select * from SYSTEM_PRIVILEGE_MAP;

```
SQL> select * from SYSTEM_PRIVILEGE_MAP;

 PRIVILEGE NAME                                         PROPERTY
---------- ---------------------------------------- ----------
        -3 ALTER SYSTEM                                      0
        -4 AUDIT SYSTEM                                      0
        -5 CREATE SESSION                                    0
        -6 ALTER SESSION                                     0
        -7 RESTRICTED SESSION                                0
       -10 CREATE TABLESPACE                                 0
       -11 ALTER TABLESPACE                                  0
       -12 MANAGE TABLESPACE                                 0
       -13 DROP TABLESPACE                                   0
       -15 UNLIMITED TABLESPACE                              0
       -20 CREATE USER                                       0

 PRIVILEGE NAME                                         PROPERTY
---------- ---------------------------------------- ----------
       -21 BECOME USER                                       0
       -22 ALTER USER                                        0
       -23 DROP USER                                         0
       -30 CREATE ROLLBACK SEGMENT                           0
       -31 ALTER ROLLBACK SEGMENT                            0
       -32 DROP ROLLBACK SEGMENT                             0
```

```
      -422 ALTER ANY PROPERTY GRAPH                          0
      -423 DROP ANY PROPERTY GRAPH                           0
      -424 READ ANY PROPERTY GRAPH                           0
      -426 CREATE TRUE CACHE                                 0
      -427 CREATE MLE                                        0
      -428 CREATE ANY MLE                                    0
      -429 ALTER ANY MLE                                     0
      -430 DROP ANY MLE                                      0
      -431 GRANT ANY SCHEMA PRIVILEGE                        0

 PRIVILEGE NAME                                         PROPERTY
---------- ---------------------------------------- ----------
      -432 ADMINISTER ROW LEVEL SECURITY POLICY              0
      -433 ADMINISTER FINE GRAINED AUDIT POLICY              0
      -434 ADMINISTER REDACTION POLICY                       0
      -435 CREATE DOMAIN                                     0
      -436 CREATE ANY DOMAIN                                 0
      -437 ALTER ANY DOMAIN                                  0
      -438 DROP ANY DOMAIN                                   0
      -439 EXECUTE ANY DOMAIN                                0

283 rows selected.

SQL>
```

Menampilkan pemetaan System Privilege (misalnya, -5 CREATE SESSION) ke ID numeriknya

SQL> create user demo identified by demo1234;

```
SQL> create user demo identified by demo1234;

User created.

SQL>
```

membuat user bernama demo dengan password demo1234

SQL> grant create session to demo;

```
SQL> grant create session to demo;

Grant succeeded.

SQL>
```

Memberikan hak akses CREATE SESSION (izin login)

SQL> grant create table to demo;

```
SQL> grant create table to demo;

Grant succeeded.

SQL>
```

Memberikan hak untuk membuat tabel

SQL> grant unlimited tablespace to demo;

```
SQL> grant unlimited tablespace to demo;

Grant succeeded.

SQL>
```

Memberikan hak untuk menggunakan tablespace tanpa batasan kuota

SQL> grant create sequence to demo;

```
SQL> grant create sequence to demo;

Grant succeeded.

SQL>
```

Memberikan hak untuk membuat sequence

SQL> grant create view to demo;

```
SQL> grant create view to demo;

Grant succeeded.

SQL>
```

Memberikan hak untuk membuat view

SQL> grant create synonym to demo;

```
SQL> grant create synonym to demo;

Grant succeeded.

SQL>
```

Memberikan hak untuk membuat sinonim

Connect to user demo

==========

```
[oracle@localhost ~]$ sqlplus

SQL*Plus: Release 23.0.0.0.0 - Production on Thu Oct 23 05:42:45 2025
Version 23.9.0.25.07

Copyright (c) 1982, 2025, Oracle.  All rights reserved.

Enter user-name: demo
Enter password:
ERROR:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/
```

```
Enter user-name: demo as sysdba
Enter password:

Connected to:
Oracle Database 23ai Free Release 23.0.0.0.0 - Develop, Learn, and Run for Free
Version 23.9.0.25.07

SQL>
```

Login pertama (user-name: demo) gagal (ORA-01017) karena tidak valid atau tidak diizinkan.

Login kedua (user-name: demo as sysdba) berhasil, ini merupakan koneksi administratif dengan privilege SYSDBA (bukan DEMO biasa)

SQL> select * from session_privs;

```
SQL> select * from session_privs;

PRIVILEGE
----------------------------------------
EXECUTE ANY DOMAIN
DROP ANY DOMAIN
ALTER ANY DOMAIN
CREATE ANY DOMAIN
CREATE DOMAIN
ADMINISTER REDACTION POLICY
ADMINISTER FINE GRAINED AUDIT POLICY
ADMINISTER ROW LEVEL SECURITY POLICY
GRANT ANY SCHEMA PRIVILEGE
DROP ANY MLE
ALTER ANY MLE

PRIVILEGE
----------------------------------------
CREATE ANY MLE
CREATE MLE
CREATE TRUE CACHE
READ ANY PROPERTY GRAPH
DROP ANY PROPERTY GRAPH
```

```
PRIVILEGE
----------------------------------------
CREATE ROLLBACK SEGMENT
DROP USER
ALTER USER
BECOME USER
CREATE USER
UNLIMITED TABLESPACE
DROP TABLESPACE
MANAGE TABLESPACE
ALTER TABLESPACE
CREATE TABLESPACE
RESTRICTED SESSION

PRIVILEGE
----------------------------------------
ALTER SESSION
CREATE SESSION
AUDIT SYSTEM
ALTER SYSTEM

279 rows selected.

SQL>
```

Menampilkan semua System Privilege saat ini, sangat banyak karena koneksi menggunakan hak SYSDBA

SQL> create table emp

(empid number constraint emp_pk primary key,

ename varchar2(100)

);

```
SQL> create table emp
( empid number constraint emp_pk primary key,
  ename varchar2(100)
);
  2    3    4
Table created.

SQL>
```

Berhasil membuat tabel EMP

SQL> insert into emp values (1,'budi');

```
SQL> insert into emp values (1,'budi');

1 row created.

SQL>
```

Memasukkan satu baris data ke dalam tabel EMP. Baris yang dimasukkan: EMPID = 1 dan ENAME = 'budi'. Perintah ini berhasil dan menghasilkan pesan: 1 row created.

SQL> select * from emp;

```
SQL> select * from emp;

     EMPID
----------
ENAME
--------------------------------------------------------------------------------
         1
budi


SQL>
```

Mengambil dan menampilkan semua kolom dan semua baris data yang ada di dalam tabel bernama EMP

SQL> alter table emp

add (salary number);

```
SQL> alter table emp
add (salary number);
  2
Table altered.

SQL>
```

Mengubah tabel EMP dengan menambahkan kolom baru bernama SALARY dengan tipe data NUMBER

SQL> select * from emp;

```
SQL> select * from emp;

     EMPID
----------
ENAME
--------------------------------------------------------------------------------
    SALARY
----------
         1
budi



SQL>
```

Mengambil dan menampilkan semua kolom dan semua baris data yang ada di dalam tabel bernama EMP

SQL> create sequence emp_s;

```
SQL> create sequence emp_s;

Sequence created.

SQL>
```

Membuat SEQUENCE baru bernama EMP_S

SQL> create index ename_ind on emp (ename);

```
SQL> create index ename_ind on emp (ename);

Index created.

SQL>
```

Membuat INDEX baru bernama ENAME_IND pada kolom ENAME di tabel EMP

SQL> create or replace view emp_v

as

select empid, ename

from emp;

```
SQL> create or replace view emp_v
as
select empid, ename
from emp;
  2    3    4
View created.

SQL>
```

Membuat atau mengganti (replace) sebuah view bernama EMP_V

SQL> alter user demo identified by demo_green;

```
SQL> alter user demo identified by demo_green;

User altered.

SQL>
```

Mengubah password dari user demo menjadi demo_green

SQL> select * from session_privs;

```
SQL> select * from session_privs;

PRIVILEGE
----------------------------------------
EXECUTE ANY DOMAIN
DROP ANY DOMAIN
ALTER ANY DOMAIN
CREATE ANY DOMAIN
CREATE DOMAIN
ADMINISTER REDACTION POLICY
ADMINISTER FINE GRAINED AUDIT POLICY
ADMINISTER ROW LEVEL SECURITY POLICY
GRANT ANY SCHEMA PRIVILEGE
DROP ANY MLE
ALTER ANY MLE

PRIVILEGE
----------------------------------------
CREATE ANY MLE
CREATE MLE
CREATE TRUE CACHE
READ ANY PROPERTY GRAPH
DROP ANY PROPERTY GRAPH
```

```
PRIVILEGE
----------------------------------------
CREATE ROLLBACK SEGMENT
DROP USER
ALTER USER
BECOME USER
CREATE USER
UNLIMITED TABLESPACE
DROP TABLESPACE
MANAGE TABLESPACE
ALTER TABLESPACE
CREATE TABLESPACE
RESTRICTED SESSION

PRIVILEGE
----------------------------------------
ALTER SESSION
CREATE SESSION
AUDIT SYSTEM
ALTER SYSTEM

279 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua System Privilege yang aktif dan dapat digunakan oleh user yang terhubung saat ini

SQL> select * from user_sys_privs;

```
SQL> select * from user_sys_privs;

USERNAME
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE ANY DOMAIN                        NO  YES YES

SYS
ADMINISTER ROW LEVEL SECURITY POLICY     NO  YES YES

SYS
DROP ANY MLE                             NO  YES YES


USERNAME
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE TRUE CACHE                        NO  YES YES
```

```
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE ROLLBACK SEGMENT                  NO  YES YES

SYS
BECOME USER                              NO  YES YES

SYS
RESTRICTED SESSION                       NO  YES YES


USERNAME
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE SESSION                           NO  YES YES


277 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua System Privilege yang diberikan kepada user yang sedang login

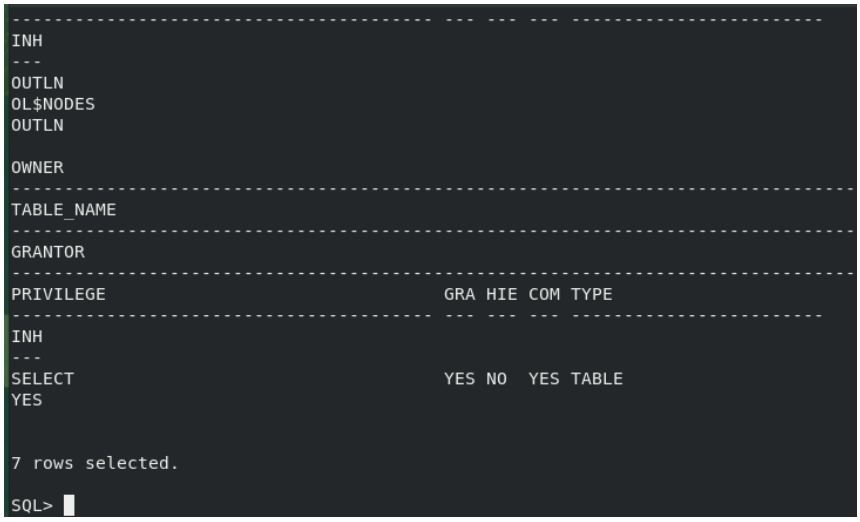
SQL> select * from user_tab_privs_recd

order by 2;

```
SQL> select * from user_tab_privs_recd
  2  order by 2;

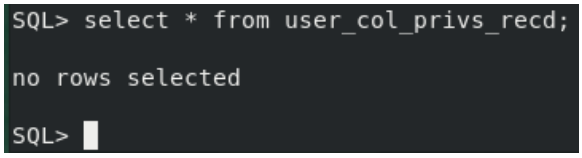
OWNER
--------------------------------------------------------------------------------
TABLE_NAME
--------------------------------------------------------------------------------
GRANTOR
--------------------------------------------------------------------------------
PRIVILEGE                                GRA HIE COM TYPE
---------------------------------------- --- --- --- -----------------------
INH
---
DVSYS
CONFIGURE_DV_INTERNAL
DVSYS

OWNER
--------------------------------------------------------------------------------
TABLE_NAME
--------------------------------------------------------------------------------
GRANTOR
```



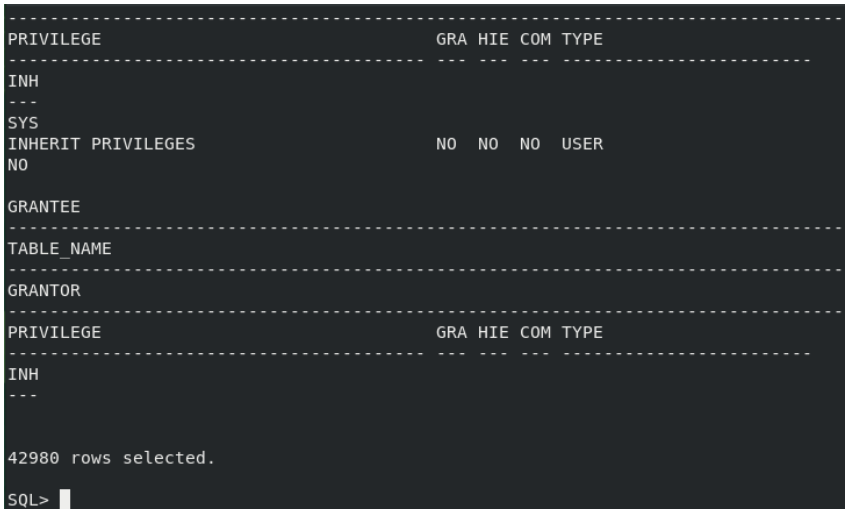
Mengambil dan menampilkan daftar hak akses yang diberikan (received) oleh user lain atau melalui role. Hasilnya diurutkan berdasarkan kolom kedua (kolom TABLE_NAME)

SQL> select * from user_col_privs_recd;



Menampilkan hak akses (seperti UPDATE atau INSERT) yang diberikan secara lengkap hanya pada kolom tertentu dalam sebuah tabel. Karena hasilnya adalah no rows selected, berarti user yang sedang terhubung belum menerima hak akses yang terbatas pada tingkat kolom manapun

SQL> select * from user_tab_privs_made;



Mengambil dan menampilkan daftar hak akses (privilege) level (tabel, view, sequence, dan lain-lain) yang telah diberikan (made) oleh user yang sedang login kepada user atau role lain

SQL> select * from user_col_privs_made;

```
SQL> select * from user_col_privs_made;

GRANTEE
--------------------------------------------------------------------------------
TABLE_NAME
--------------------------------------------------------------------------------
COLUMN_NAME
--------------------------------------------------------------------------------
GRANTOR
--------------------------------------------------------------------------------
PRIVILEGE                                GRA COM INH
---------------------------------------- --- --- ---
IMP_FULL_DATABASE
KET$_CLIENT_CONFIG
FIELD_3

GRANTEE
--------------------------------------------------------------------------------
TABLE_NAME
--------------------------------------------------------------------------------
COLUMN_NAME
--------------------------------------------------------------------------------
GRANTOR
--------------------------------------------------------------------------------
```

```
--------------------------------------------------------------------------------
GRANTOR
--------------------------------------------------------------------------------
PRIVILEGE                                GRA COM INH
---------------------------------------- --- --- ---
IMP_FULL_DATABASE
KET$_CLIENT_CONFIG
FIELD_1

GRANTEE
--------------------------------------------------------------------------------
TABLE_NAME
--------------------------------------------------------------------------------
COLUMN_NAME
--------------------------------------------------------------------------------
GRANTOR
--------------------------------------------------------------------------------
PRIVILEGE                                GRA COM INH
---------------------------------------- --- --- ---
SYS
UPDATE                                   NO  YES YES


SQL>
```

Mengambil dan menampilkan daftar hak akses (privilege) level kolom yang telah diberikan (made) oleh user yang sedang login kepada user atau role lain

==========

```
[oracle@localhost ~]$ sqlplus

SQL*Plus: Release 23.0.0.0.0 - Production on Thu Oct 23 06:05:39 2025
Version 23.9.0.25.07

Copyright (c) 1982, 2025, Oracle.  All rights reserved.

Enter user-name: sys as sysdba
Enter password:

Connected to:
Oracle Database 23ai Free Release 23.0.0.0.0 - Develop, Learn, and Run for Free
Version 23.9.0.25.07

SQL>
```

Exit, lalu masuk lagi dengan sqlplus dengan username sys as sysdba dan password oracle

SQL> create role manager;

```
SQL> create role manager;

Role created.

SQL>
```

Membuat ROLE baru bernama MANAGER

SQL> grant create table, create view, create sequence

to manager;

```
SQL> grant create table, create view, create sequence
to manager;
  2
Grant succeeded.

SQL>
```

Setiap user yang memiliki (grant) role MANAGER, secara otomatis akan mendapatkan hak untuk CREATE TABLE, CREATE VIEW, dan CREATE SEQUENCE

SQL> SELECT * FROM ROLE_SYS_PRIVS

WHERE ROLE='MANAGER';

```
SQL> SELECT * FROM ROLE_SYS_PRIVS
WHERE ROLE='MANAGER';
  2
ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
MANAGER
CREATE SEQUENCE                          NO  NO  NO

MANAGER
CREATE VIEW                              NO  NO  NO

MANAGER
CREATE TABLE                             NO  NO  NO


SQL>
```

Mengambil dan menampilkan semua System Privilege yang telah diberikan kepada Role bernama MANAGER

SQL> CREATE USER budi identified by budi123;

```
SQL> CREATE USER budi identified by budi123;

User created.

SQL>
```

Membuat User baru bernama BUDI dengan password budi123.

SQL> grant create session to budi;

```
SQL> grant create session to budi;

Grant succeeded.

SQL>
```

Memberikan System Privilege CREATE SESSION ke user BUDI

SQL> grant unlimited tablespace to budi;

```
SQL> grant unlimited tablespace to budi;

Grant succeeded.

SQL>
```

Memberikan System Privilege UNLIMITED TABLESPACE kepada user BUDI

SQL> grant manager to budi;

```
SQL> grant manager to budi;

Grant succeeded.

SQL>
```

Memberikan Role bernama MANAGER kepada user BUDI

SQL> CREATE ROLE QONLY;

```
SQL> CREATE ROLE QONLY;

Role created.

SQL>
```

Membuat ROLE baru bernama QONLY (Query Only)

SQL> GRANT SELECT ANY TABLE TO QONLY;

```
SQL> GRANT SELECT ANY TABLE TO QONLY;

Grant succeeded.

SQL>
```

Memberikan System Privilege SELECT ANY TABLE ke Role bernama QONLY

SQL> GRANT QONLY TO BUDI;

```
SQL> GRANT QONLY TO BUDI;

Grant succeeded.

SQL>
```

Memberikan Role bernama QONLY kepada user BUDI

SQL> CREATE ROLE IUD_EMP;

```
SQL> CREATE ROLE IUD_EMP;

Role created.

SQL>
```

Membuat ROLE baru bernama IUD_EMP

SQL> GRANT IUD_EMP TO BUDI;

```
SQL> GRANT IUD_EMP TO BUDI;

Grant succeeded.

SQL>
```

Memberikan Role bernama IUD_EMP kepada user BUDI

SQL> SELECT * FROM ROLE_TAB_PRIVS

WHERE ROLE='IUD_EMP';

```
SQL> SELECT * FROM ROLE_TAB_PRIVS
WHERE ROLE='IUD_EMP';
  2
no rows selected

SQL>
```

Mengambil dan menampilkan daftar Object Privilege (seperti INSERT, UPDATE, DELETE) yang telah diberikan kepada Role bernama IUD_EMP

SQL> select * from session_privs;

```
SQL> select * from session_privs;

PRIVILEGE
----------------------------------------
EXECUTE ANY DOMAIN
DROP ANY DOMAIN
ALTER ANY DOMAIN
CREATE ANY DOMAIN
CREATE DOMAIN
ADMINISTER REDACTION POLICY
ADMINISTER FINE GRAINED AUDIT POLICY
ADMINISTER ROW LEVEL SECURITY POLICY
GRANT ANY SCHEMA PRIVILEGE
DROP ANY MLE
ALTER ANY MLE

PRIVILEGE
----------------------------------------
CREATE ANY MLE
CREATE MLE
CREATE TRUE CACHE
READ ANY PROPERTY GRAPH
DROP ANY PROPERTY GRAPH
```

```
PRIVILEGE
----------------------------------------
CREATE ROLLBACK SEGMENT
DROP USER
ALTER USER
BECOME USER
CREATE USER
UNLIMITED TABLESPACE
DROP TABLESPACE
MANAGE TABLESPACE
ALTER TABLESPACE
CREATE TABLESPACE
RESTRICTED SESSION

PRIVILEGE
----------------------------------------
ALTER SESSION
CREATE SESSION
AUDIT SYSTEM
ALTER SYSTEM

279 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua System Privilege yang aktif dan dapat digunakan oleh user yang terhubung saat ini

SQL> select * from user_sys_privs;

```
SQL> select * from user_sys_privs;

USERNAME
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE ANY DOMAIN                        NO  YES YES

SYS
ADMINISTER ROW LEVEL SECURITY POLICY     NO  YES YES

SYS
DROP ANY MLE                             NO  YES YES


USERNAME
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE TRUE CACHE                        NO  YES YES
```

```
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE ROLLBACK SEGMENT                  NO  YES YES

SYS
BECOME USER                              NO  YES YES

SYS
RESTRICTED SESSION                       NO  YES YES


USERNAME
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SYS
CREATE SESSION                           NO  YES YES


277 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua System Privilege yang diberikan kepada user yang sedang login

SQL> select * from user_role_privs;

```
SQL> select * from user_role_privs;

USERNAME
--------------------------------------------------------------------------------
GRANTED_ROLE
--------------------------------------------------------------------------------
ADM DEL DEF OS_ COM INH
--- --- --- --- --- ---
SYS
MANAGER
YES NO  YES NO  NO  NO

SYS
QONLY
YES NO  YES NO  NO  NO

USERNAME
--------------------------------------------------------------------------------
GRANTED_ROLE
--------------------------------------------------------------------------------
ADM DEL DEF OS_ COM INH
```

```
ADM DEL DEF OS_ COM INH
--- --- --- --- --- ---
PGX_SESSION_NEW_GRAPH
YES NO  YES NO  YES YES

SYS
PGX_SESSION_SET_IDLE_TIMEOUT
YES NO  YES NO  YES YES


USERNAME
--------------------------------------------------------------------------------
GRANTED_ROLE
--------------------------------------------------------------------------------
ADM DEL DEF OS_ COM INH
--- --- --- --- --- ---
SYS
OLAP_XS_ADMIN
YES NO  YES NO  YES YES


106 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua Role (peran) yang diberikan kepada user yang sedang login

SQL> select * from role_sys_privs;

```
SQL> select * from role_sys_privs;

ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
PDB_DBA
CREATE PLUGGABLE DATABASE                NO  NO  NO

PDB_DBA
CREATE SESSION                           NO  NO  NO

MANAGER
CREATE SEQUENCE                          NO  NO  NO


ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
MANAGER
CREATE VIEW                              NO  NO  NO
```

```
XSTREAM_APPLY
DEQUEUE ANY QUEUE                        YES YES YES

OEM_MONITOR
MANAGE ANY QUEUE                         NO  YES YES

OEM_MONITOR
CREATE SESSION                           NO  YES YES


ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
DB_DEVELOPER_ROLE
CREATE CUBE                              NO  YES YES

OLAP_USER
CREATE MEASURE FOLDER                    NO  YES YES


548 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua System Privilege yang ada di semua Role di PDB

SQL> SELECT * FROM ROLE_TAB_PRIVS

WHERE ROLE='IUD_EMP';

```
SQL> SELECT * FROM ROLE_TAB_PRIVS
WHERE ROLE='IUD_EMP';
  2
no rows selected

SQL>
```

View ROLE_TAB_PRIVS untuk melihat hak-hak yang dimiliki role tertentu (bukan hak sistem). Hasilnya no rows selected, berarti meskipun role IUD_EMP telah dibuat dan diberikan kepada user lain, role belum diberikan hak akses pada tabel manapun

SQL> create table student

(student_id number,

  student_name varchar2(100)

);

```
SQL> create table student
( student_id number,
  student_name varchar2(100)
);
  2    3    4
Table created.

SQL>
```

Membuat Tabel baru bernama STUDENT

SQL> grant select on

student to public;

```
SQL> grant select on
student to public;
  2
Grant succeeded.

SQL>
```

Memberikan Object Privilege SELECT pada tabel STUDENT kepada PUBLIC

==========

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> select * from dba_users;

```
------------------------------ ------------------------------
LOCAL_TEMP_TABLESPACE          CREATED
------------------------------ ---------
PROFILE
--------------------------------------------------------------------------------
INITIAL_RSRC_CONSUMER_GROUP
--------------------------------------------------------------------------------
EXTERNAL_NAME
--------------------------------------------------------------------------------
PASSWORD_VERSIONS E AUTHENTI P COM
---------------- - -------- - ---
LAST_LOGIN                                                                    0
------------------------------------------------------------------------------ -
INH
---
DEFAULT_COLLATION
--------------------------------------------------------------------------------
IMP ALL EXT PASSWORD_ MAN PRO REA DIC CLO APP MFA
--- --- --- --------- --- --- --- --- --- --- ---------


38 rows selected.

SQL>
```

Mengambil dan menampilkan semua kolom dan semua baris dari view sistem DBA_USERS

SQL> create user jane identified by jane;

```
SQL> alter session set container=cdb$root;

Session altered.

SQL> create user jane identified by jane;
create user jane identified by jane
*
ERROR at line 1:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/
```

Error karena berada di ke cdb$root, seharusnya di freepdb1

```
SQL> create user jane identified by jane;

User created.

SQL>
```

membuat User baru bernama JANE dengan password jane

SQL> show parameter common_user_prefix;

```
SQL> show parameter common_user_prefix;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
common_user_prefix                   string
SQL>
```

Menampilkan nilai dari parameter COMMON_USER_PREFIX

SQL> create user c##jane identified by jane container=all;

```
SQL> create user c##jane identified by jane container=all;
create user c##jane identified by jane container=all
*
ERROR at line 1:
ORA-65050: Common DDLs only allowed in root.
Help: https://docs.oracle.com/error-help/db/ora-65050/


SQL>
```

Error karena Common (CONTAINER=ALL) hanya boleh dijalankan ketika terhubung ke CDB$ROOT, bukan dari Pluggable Database (PDB) seperti FREEPDB1

SQL> grant create session to c##jane ;

```
SQL> grant create session to c##jane ;
grant create session to c##jane
                        *
ERROR at line 1:
ORA-01917: user or role 'C##JANE' does not exist
Help: https://docs.oracle.com/error-help/db/ora-01917/


SQL>
```

Error karena C##JANE tidak ditemukan di container sekarang. User C##JANE sebelumnya gagal dibuat (karena harus dilakukan di CDB$ROOT), sehingga user tersebut tidak pernah ada

SQL> grant create session to c##jane container=all;

```
SQL> grant create session to c##jane container=all;
grant create session to c##jane container=all
                        *
ERROR at line 1:
ORA-01917: user or role 'C##JANE' does not exist
Help: https://docs.oracle.com/error-help/db/ora-01917/


SQL>
```

Error karena user atau role bernama C##JANE tidak ditemukan. C##JANE gagal karena administrator mencoba membuat Common User dari dalam PDB (yang seharusnya dari CDB$ROOT), sehingga user tidak pernah terdaftar

==========

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> select * from dba_roles;

```
SQL> select * from dba_roles;

ROLE
--------------------------------------------------------------------------------
   ROLE_ID PASSWORD AUTHENTICAT COM O INH IMP
---------- -------- ----------- --- - --- ---
EXTERNAL_NAME
--------------------------------------------------------------------------------
APP
---
CONNECT
         2 NO       NONE        YES Y YES NO

NO


ROLE
--------------------------------------------------------------------------------
   ROLE_ID PASSWORD AUTHENTICAT COM O INH IMP
---------- -------- ----------- --- - --- ---
EXTERNAL_NAME
--------------------------------------------------------------------------------
APP
```

```
---
QONLY
       139 NO       NONE        NO  N NO  NO

NO


ROLE
--------------------------------------------------------------------------------
   ROLE_ID PASSWORD AUTHENTICAT COM O INH IMP
---------- -------- ----------- --- - --- ---
EXTERNAL_NAME
--------------------------------------------------------------------------------
APP
---
IUD_EMP
       140 NO       NONE        NO  N NO  NO

NO


121 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua Role yang ada di container sekarang

SQL> create role test_role;

```
SQL> create role test_role;

Role created.

SQL>
```

Membuat ROLE baru bernama TEST_ROLE

SQL> show parameter common_user_prefix;

```
SQL> show parameter common_user_prefix;

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
common_user_prefix                   string
SQL>
```

Menampilkan nilai dari parameter inisialisasi COMMON_USER_PREFIX

SQL> create role c##test_role container=all;

```
SQL> create role c##test_role container=all;
create role c##test_role container=all
*
ERROR at line 1:
ORA-65050: Common DDLs only allowed in root.
Help: https://docs.oracle.com/error-help/db/ora-65050/


SQL>
```

Membuat Common Role (C##TEST_ROLE) yang berlaku di semua PDB (container=all). Error karena ORA-65050: Common DDLs only allowed in root. Pembuatan objek yang berlaku di seluruh container hanya boleh dilakukan dari CDB$ROOT, bukan dari PDB tempat yang saat ini terhubung

SQL> grant create session to c##test_role container=all;

```
SQL> grant create session to c##test_role container=all;
grant create session to c##test_role container=all
                        *
ERROR at line 1:
ORA-01917: user or role 'C##TEST_ROLE' does not exist
Help: https://docs.oracle.com/error-help/db/ora-01917/


SQL>
```

Memberikan hak CREATE SESSION ke Common Role C##TEST_ROLE dan menetapkan hak tersebut di semua container (container=all). Gagal karena ORA-01917: user or role 'C##TEST_ROLE' does not exist, karena sebelumnya cara untuk membuat role C##TEST_ROLE sebelumnya juga gagal (harus dijalankan dari CDB$ROOT), sehingga role tersebut tidak terdaftar di database

SQL> grant create table to c##test_role container=all;

```
SQL> grant create table to c##test_role container=all;
grant create table to c##test_role container=all
                      *
ERROR at line 1:
ORA-01917: user or role 'C##TEST_ROLE' does not exist
Help: https://docs.oracle.com/error-help/db/ora-01917/


SQL>
```

Memberikan hak CREATE TABLE ke Common Role C##TEST_ROLE dan menetapkan hak tersebut berlaku di semua container (container=all). Gagal karena ORA-01917: user or role 'C##TEST_ROLE' does not exist, karena sebelumnya untuk membuat role C##TEST_ROLE gagal (harus dijalankan dari CDB$ROOT), sehingga role tersebut tidak terdaftar di database

SQL> create user c##dark identified by dark;

```
SQL> create user c##dark identified by dark;
create user c##dark identified by dark
            *
ERROR at line 1:
ORA-65094: invalid local user or role name
Help: https://docs.oracle.com/error-help/db/ora-65094/


SQL>
```

Membuat user c##dark dengan password dark. Karena menggunakan prefix c## tetapi tidak memakai CONTAINER=ALL, database mencoba membuatnya sebagai Local User di PDB (FREEPDB1). Gagal karena ORA-65094: invalid local user or role name. Di dalam PDB, Local User tidak boleh menggunakan prefix C##. Prefix C## hanya diperbolehkan untuk Common User atau Common Role, yang harus dibuat dengan CONTAINER=ALL (harus dijalankan dari CDB$ROOT)

SQL> grant c##test_role to c##dark container=all;

```
SQL> grant c##test_role to c##dark container=all;
grant c##test_role to c##dark container=all
                      *
ERROR at line 1:
ORA-01917: user or role 'C##DARK' does not exist
Help: https://docs.oracle.com/error-help/db/ora-01917/


SQL>
```

Memberikan Common Role C##TEST_ROLE ke Common User C##DARK dan menetapkan pemberian ini berlaku di semua container (container=all). Error karena ORA-01917: user or role 'C##DARK' does not exist. Error karena user C##DARK tidak ditemukan di database. Sebelumnya, cara untuk membuat C##DARK gagal karena mencoba menggunakan prefix C## sebagai user lokal di PDB (yang seharusnya dibuat dari CDB$ROOT dengan CONTAINER=ALL)

SQL> create user c##kim identified by kim;

```
SQL> create user c##kim identified by kim;
create user c##kim identified by kim
            *
ERROR at line 1:
ORA-65094: invalid local user or role name
Help: https://docs.oracle.com/error-help/db/ora-65094/


SQL>
```

membuat user c##kim dengan password kim. Karena user menggunakan prefix c## tetapi tidak menggunakan CONTAINER=ALL, database mencoba membuatnya sebagai Local User di PDB (FREEPDB1). Gagal karena ORA-65094: invalid local user or role name. Di dalam PDB, Local User tidak boleh menggunakan prefix C##. Prefix C## untuk Common User atau Common Role, yang seharusnya dibuat dengan menggunakan CONTAINER=ALL dan dijalankan dari CDB$ROOT

SQL> alter session set container=freepdb1;

```
SQL> alter session set container=freepdb1;

Session altered.

SQL>
```

Mengubah ke freepdb1

SQL> grant c##test_role to c##kim;

```
SQL> grant c##test_role to c##kim;
grant c##test_role to c##kim
      *
ERROR at line 1:
ORA-01919: Role 'C##TEST_ROLE' does not exist.
Help: https://docs.oracle.com/error-help/db/ora-01919/


SQL>
```

memberikan Role C##TEST_ROLE ke User C##KIM. Gagal karena ORA-01919: Role 'C##TEST_ROLE' does not exist. Error karena role C##TEST_ROLE tidak ditemukan di database atau container sekarang. Terjadi karena sebelumnya untuk membuat role C##TEST_ROLE (Common Role) gagal karena harus dilakukan dari CDB$ROOT, bukan dari PDB

==========

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> show user;

```
SQL> show user;
USER is "SYS"
SQL>
```

Menampilkan nama user yang sekarang terhubung dan sedang digunakan

SQL> select * from session_privs;

```
SQL> select * from session_privs;

PRIVILEGE
----------------------------------------
EXECUTE ANY DOMAIN
DROP ANY DOMAIN
ALTER ANY DOMAIN
CREATE ANY DOMAIN
CREATE DOMAIN
ADMINISTER REDACTION POLICY
ADMINISTER FINE GRAINED AUDIT POLICY
ADMINISTER ROW LEVEL SECURITY POLICY
GRANT ANY SCHEMA PRIVILEGE
DROP ANY MLE
ALTER ANY MLE

PRIVILEGE
----------------------------------------
CREATE ANY MLE
CREATE MLE
CREATE TRUE CACHE
```

```
PRIVILEGE
----------------------------------------
CREATE ROLLBACK SEGMENT
DROP USER
ALTER USER
BECOME USER
CREATE USER
UNLIMITED TABLESPACE
DROP TABLESPACE
MANAGE TABLESPACE
ALTER TABLESPACE
CREATE TABLESPACE
RESTRICTED SESSION

PRIVILEGE
----------------------------------------
ALTER SESSION
CREATE SESSION
AUDIT SYSTEM
ALTER SYSTEM

279 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua System Privilege yang aktif dan dapat digunakan oleh user yang terhubung sekarang

SQL> show con_id;

```
SQL> show con_id;

CON_ID
------------------------------
3
SQL>
```

Menampilkan ID numerik dari container database sekarang yang sedang terhubung

SQL> select * from V$TABLESPACE

where con_id=1;

```
SQL> select * from V$TABLESPACE
where con_id=1;
  2
no rows selected

SQL>
```

Menampilkan semua TABLESPACE milik CDB$ROOT (root container, karena CON_ID=1). Gagal dengan hasil no rows selected, karena saat ini terhubung ke PDB (CON_ID > 1), dan sesi PDB tidak dapat melihat metadata yang secara lengkap yang dimiliki oleh CDB$ROOT

SQL> create user c##copysis identified by copysis

default tablespace users temporary tablespace temp account unlock;

```
SQL> create user c##copysis identified by copysis
  2  default tablespace users temporary tablespace temp account unlock;
create user c##copysis identified by copysis
            *
ERROR at line 1:
ORA-65094: invalid local user or role name
Help: https://docs.oracle.com/error-help/db/ora-65094/


SQL>
```

membuat user c##copysis dengan menetapkan default tablespace, temporary tablespace, dan membuka akun (unlock). Karena user menggunakan prefix c## tetapi tidak menggunakan CONTAINER=ALL, database mencoba membuatnya sebagai Local User di PDB (FREEPDB1). Gagal karena ORA-65094: invalid local user or role name. Di dalam PDB, Local User tidak boleh menggunakan prefix C##. Prefix C## hanya untuk Common User atau Common Role

SQL> grant create session, dba,sysdba to c##copysis container=all;

```
SQL> grant create session, dba,sysdba to c##copysis container=all;
grant create session, dba,sysdba to c##copysis container=all
                                    *
ERROR at line 1:
ORA-01917: user or role 'C##COPYSIS' does not exist
Help: https://docs.oracle.com/error-help/db/ora-01917/


SQL>
```

memberikan hak CREATE SESSION, Role DBA, dan SYSDBA ke Common User C##COPYSIS, dengan menetapkan hak tersebut berlaku di semua container (container=all). Gagal karena ORA-01917: user or role 'C##COPYSIS' does not exist. Error karena user C##COPYSIS tidak ditemukan di database. Cara sebelumnya untuk membuat user ini gagal karena mencoba menggunakan prefix C## sebagai user lokal di PDB, sehingga user tersebut tidak pernah terdaftar

SQL> select * from dba_users

where username=upper('c##copysis');

```
SQL> select * from dba_users
where username=upper('c##copysis');
  2
no rows selected

SQL>
```

mengambil semua detail dari view DBA_USERS untuk user bernama C##COPYSIS. Nama user diubah menjadi huruf kapital (UPPER()) karena menyimpan user dan objek dalam case-insensitive query secara default menggunakan huruf besar. Hasilnya no rows selected. User bernama C##COPYSIS tidak

terdaftar dalam container database saat ini (PDB FREEPDB1). Sesuai dengan error sebelumnya, di mana untuk membuat C##COPYSIS gagal karena melanggar aturan penamaan Local User di PDB

SQL> SELECT * FROM SESSION_PRIVS;

```
SQL> select * from session_privs;

PRIVILEGE
----------------------------------------
EXECUTE ANY DOMAIN
DROP ANY DOMAIN
ALTER ANY DOMAIN
CREATE ANY DOMAIN
CREATE DOMAIN
ADMINISTER REDACTION POLICY
ADMINISTER FINE GRAINED AUDIT POLICY
ADMINISTER ROW LEVEL SECURITY POLICY
GRANT ANY SCHEMA PRIVILEGE
DROP ANY MLE
ALTER ANY MLE

PRIVILEGE
----------------------------------------
CREATE ANY MLE
CREATE MLE
CREATE TRUE CACHE
```

```
PRIVILEGE
----------------------------------------
CREATE ROLLBACK SEGMENT
DROP USER
ALTER USER
BECOME USER
CREATE USER
UNLIMITED TABLESPACE
DROP TABLESPACE
MANAGE TABLESPACE
ALTER TABLESPACE
CREATE TABLESPACE
RESTRICTED SESSION

PRIVILEGE
----------------------------------------
ALTER SESSION
CREATE SESSION
AUDIT SYSTEM
ALTER SYSTEM

279 rows selected.

SQL>
```

Mengambil dan menampilkan daftar lengkap semua System Privilege dan bisa digunakan oleh user yang terhubung

SQL> select * from role_sys_privs

where role='DBA';

```
SQL> select * from role_sys_privs
where role='DBA';
  2
ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
DBA
DROP ANY DOMAIN                          NO  YES YES

DBA
ADMINISTER REDACTION POLICY              NO  YES YES

DBA
ADMINISTER FINE GRAINED AUDIT POLICY     NO  YES YES


ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
DBA
```

```
DBA
ALTER ANY ROLE                           NO  YES YES

DBA
GRANT ANY ROLE                           NO  YES YES

DBA
CREATE SEQUENCE                          NO  YES YES


ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
DBA
CREATE ANY TABLE                         NO  YES YES

DBA
MANAGE TABLESPACE                        NO  YES YES


257 rows selected.

SQL>
```

Mengambil dan menampilkan daftar semua System Privilege yang ada di dalam Role bernama DBA

SQL> SELECT * FROM SESSION_PRIVS

WHERE PRIVILEGE NOT IN (select PRIVILEGE from role_sys_privs where role='DBA');

```
SQL> SELECT * FROM SESSION_PRIVS
WHERE PRIVILEGE NOT IN (select PRIVILEGE from role_sys_privs where role='DBA' );
  2
PRIVILEGE
----------------------------------------
GRANT ANY SCHEMA PRIVILEGE
CREATE TRUE CACHE
TABLE RETENTION
ADMINISTER SQL FIREWALL
TEXT DATASTORE ACCESS
WRITE ANY ANALYTIC VIEW CACHE
READ ANY ANALYTIC VIEW CACHE
INHERIT ANY REMOTE PRIVILEGES
TRANSLATE ANY SQL
INHERIT ANY PRIVILEGES
EXEMPT REDACTION POLICY

PRIVILEGE
----------------------------------------
PURGE DBA_RECYCLEBIN
KEEP SYSGUID
KEEP DATE TIME
```

```
WRITE ANY ANALYTIC VIEW CACHE
READ ANY ANALYTIC VIEW CACHE
INHERIT ANY REMOTE PRIVILEGES
TRANSLATE ANY SQL
INHERIT ANY PRIVILEGES
EXEMPT REDACTION POLICY

PRIVILEGE
----------------------------------------
PURGE DBA_RECYCLEBIN
KEEP SYSGUID
KEEP DATE TIME
ADMINISTER KEY MANAGEMENT
ALTER DATABASE LINK
ALTER PUBLIC DATABASE LINK
EXEMPT IDENTITY POLICY
EXEMPT ACCESS POLICY
SYSOPER
SYSDBA
UNLIMITED TABLESPACE

22 rows selected.

SQL>
```

Mengambil dan menampilkan System Privilege, kecuali hak yang ada dalam Role DBA

==========

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> show user;

```
SQL> show user;
USER is "SYS"
SQL>
```

Menampilkan nama user yang sedang terhubung dan sedang digunakan sekarang

SQL> alter session set container=freepdb1;

```
SQL> alter session set container=freepdb1;

Session altered.

SQL>
```

Mengubah container ke freepdb1

SQL> show con_name;

```
SQL> show con_name;


CON_NAME
------------------------------
FREEPDB1
SQL> SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> show con_id;

```
SQL> SQL> show con_id;

CON_ID
------------------------------
3
SQL>
```

Menampilkan ID numerik dari container database sekarang yang sedang terhubung

SQL> select * from V$TABLESPACE;

```
SQL> select * from V$TABLESPACE;

       TS# NAME                           INC BIG FLA ENC     CON_ID
---------- ------------------------------ --- --- --- --- ----------
         0 SYSTEM                         YES YES YES              3
         1 SYSAUX                         YES YES YES              3
         2 UNDOTBS1                       YES YES YES              3
         3 TEMP                           NO  NO  YES              3
         6 USERS                          YES YES YES              3

SQL>
```

Mengambil dan menampilkan semua kolom dan semua baris dari view tablespace

SQL> create user freepdb1_admin identified by freepdb1

default tablespace users temporary tablespace temp account unlock;

```
SQL> create user freepdb1_admin identified by freepdb1
default tablespace users temporary tablespace temp account unlock;
  2
User created.

SQL>
```

Membuat User Lokal baru FREEPDB1_ADMIN dengan konfigurasi tablespace dan status akun tertentu

SQL> grant create session, dba to freepdb1_admin;

```
SQL> grant create session, dba to freepdb1_admin;

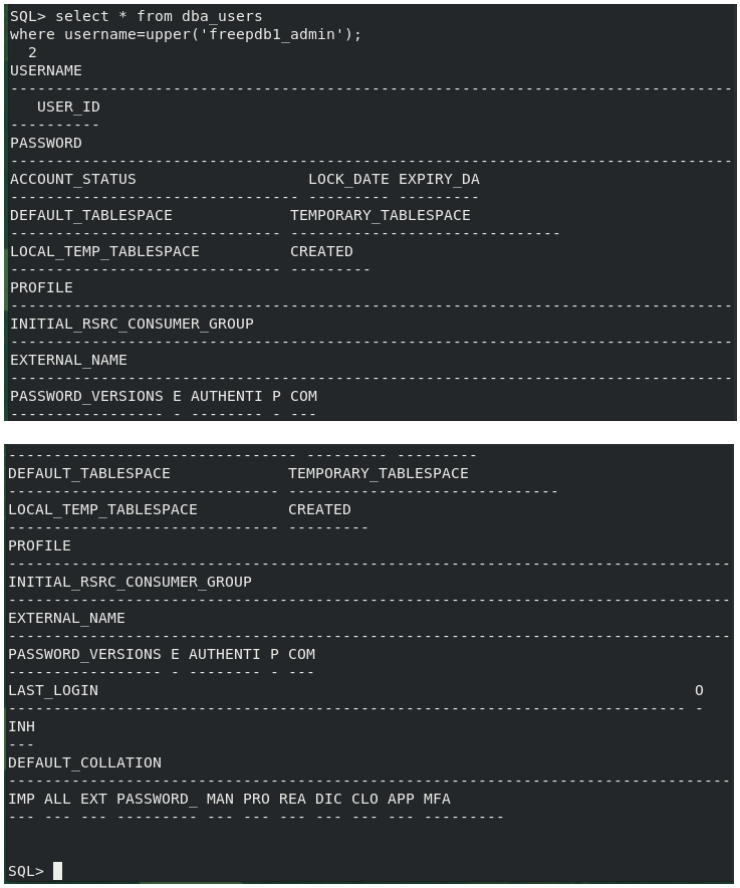
Grant succeeded.

SQL>
```

Memberikan hak CREATE SESSION dan Role DBA kepada user FREEPDB1_ADMIN

SQL> select * from dba_users

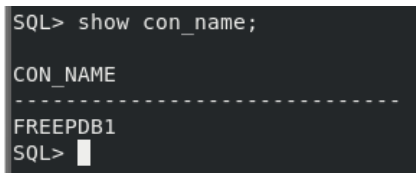
where username=upper('freepdb1_admin');



Mengambil dan menampilkan semua detail dari view DBA_USERS untuk user FREEPDB1_ADMIN

==========

SQL> show con_name;



Menampilkan nama container saat ini: FREEPDB1

SQL> show user;

```
SQL> show user;
USER is "SYS"
SQL>
```

Menampilkan nama user yang sedang terhubung dan sedang digunakan sekarang

SQL> alter session set container=freepdb1;

```
SQL> alter session set container=freepdb1;

Session altered.

SQL>
```

Mengubah container ke freepdb1

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> create role master_role;

```
SQL> create role master_role;

Role created.

SQL>
```

Membuat ROLE baru bernama MASTER_ROLE

SQL> grant create session to master_role;

```
SQL> grant create session to master_role;

Grant succeeded.

SQL>
```

Memberikan System Privilege CREATE SESSION ke Role bernama MASTER_ROLE

SQL> grant create table to master_role;

```
SQL> grant create table to master_role;

Grant succeeded.

SQL>
```

Memberikan System Privilege CREATE TABLE kepada Role bernama MASTER_ROLE

SQL> SELECT * FROM ROLE_SYS_PRIVS

where role= upper('master_role');

```
SQL> SELECT * FROM ROLE_SYS_PRIVS
where role= upper('master_role');
  2
ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
MASTER_ROLE
CREATE TABLE                             NO  NO  NO

MASTER_ROLE
CREATE SESSION                           NO  NO  NO


SQL>
```

Mengambil dan menampilkan semua System Privilege yang diberikan ke Role bernama MASTER_ROLE

SQL> create role sub_master_role;

```
SQL> create role sub_master_role;

Role created.

SQL>
```

Membuat **ROLE** baru bernama **SUB_MASTER_ROLE**

SQL> grant create view to sub_master_role;

```
SQL> grant create view to sub_master_role;

Grant succeeded.

SQL>
```

memberikan System Privilege CREATE VIEW ke Role bernama SUB_MASTER_ROLE

SQL> grant sub_master_role to master_role;

```
SQL> grant sub_master_role to master_role;

Grant succeeded.

SQL>
```

Memberikan Role SUB_MASTER_ROLE ke Role MASTER_ROLE

SQL> SELECT * FROM ROLE_SYS_PRIVS

where role= upper('master_role');

```
SQL> SELECT * FROM ROLE_SYS_PRIVS
where role= upper('master_role');
  2
ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
MASTER_ROLE
CREATE TABLE                             NO  NO  NO

MASTER_ROLE
CREATE SESSION                           NO  NO  NO


SQL>
```

Mengambil dan menampilkan daftar semua System Privilege yang diberikan ke Role bernama MASTER_ROLE dimana Role huruf kapital

SQL> select * from DBA_role_PRIVS

where GRANTEE=upper('master_role');

```
SQL> select * from DBA_role_PRIVS
where GRANTEE=upper('master_role');
  2
GRANTEE
--------------------------------------------------------------------------------
GRANTED_ROLE
--------------------------------------------------------------------------------
ADM DEL DEF COM INH
--- --- --- --- ---
MASTER_ROLE
SUB_MASTER_ROLE
NO  NO  YES NO  NO


SQL>
```

Mengambil dan menampilkan daftar Role yang diberikan ke Role bernama MASTER_ROLE dimana Role huruf kapital

SQL> SELECT * FROM ROLE_SYS_PRIVS

where role= upper('SUB_MASTER_ROLE');

```
SQL> SELECT * FROM ROLE_SYS_PRIVS
where role= upper('SUB_MASTER_ROLE');
  2
ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
SUB_MASTER_ROLE
CREATE VIEW                              NO  NO  NO


SQL>
```

Mengambil dan menampilkan daftar semua System Privilege yang telah diberikan secara langsung kepada Role bernama SUB_MASTER_ROLE dimana Role huruf kapital

SQL> create user bd111 identified by bd111;

```
SQL> create user bd111 identified by bd111;

User created.

SQL>
```

Membuat User Lokal baru bernama BD111 dengan password bd111

SQL> grant master_role to bd111;

```
SQL> grant master_role to bd111;

Grant succeeded.

SQL>
```

Memberikan Role bernama MASTER_ROLE kepada user BD111

Connect to new user

==========

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> show user;

```
SQL> show user;
USER is "SYS"
SQL>
```

Menampilkan nama user yang sedang terhubung dan sedang digunakan sekarang

SQL> SELECT * FROM CDB_USERS

WHERE USERNAME='PDB10_ADMIN';

```
SQL> SELECT * FROM CDB_USERS
WHERE USERNAME='PDB10_ADMIN';
  2
no rows selected

SQL>
```

Mengambil dan menampilkan semua user dari view CDB_USERS yang nama usernya cocok dengan PDB10_ADMIN. Hasilnya no rows selected, berarti user bernama PDB10_ADMIN tidak ditemukan di container database manapun (CDB Root maupun PDB lain) karena user belum dibuat

SQL> select * from CDB_SYS_PRIVS

where GRANTEE='PDB10_ADMIN';

```
SQL> select * from CDB_SYS_PRIVS
where GRANTEE='PDB10_ADMIN';
  2
no rows selected

SQL>
```

Mengambil dan menampilkan semua System Privilege yang diberikan kepada user atau role bernama PDB10_ADMIN. Hasilnya no rows selected, karena user atau role bernama PDB10_ADMIN belum dibuat

SQL> select * from CDB_role_PRIVS

where GRANTEE='PDB10_ADMIN';

```
SQL> select * from CDB_role_PRIVS
where GRANTEE='PDB10_ADMIN';
  2
no rows selected

SQL>
```

Mengambil dan menampilkan daftar Role yang telah diberikan ke user atau role bernama PDB10_ADMIN. Hasilnya no rows selected, berarti user atau role bernama PDB10_ADMIN saat ini belum diberikan role apa pun yang terdaftar karena belum dibuat

SQL> ALTER SESSION SET CONTAINER=PDB10;

```
SQL> ALTER SESSION SET CONTAINER=PDB10;
ERROR:
ORA-65011: Pluggable database PDB10 does not exist.
Help: https://docs.oracle.com/error-help/db/ora-65011/


SQL>
```

mengubah dari container saat ini ke PDB10. Error karena ORA-65011: Pluggable database PDB10 does not exist. Error karena database tidak dapat menemukan PDB10 karena belum dibuat

SQL> SHOW CON_NAME;

```
SQL> SHOW CON_NAME;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> SELECT * FROM ROLE_SYS_PRIVS

WHERE ROLE='PDB_DBA';

```
SQL> SELECT * FROM ROLE_SYS_PRIVS
WHERE ROLE='PDB_DBA';
  2
ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
PDB_DBA
CREATE PLUGGABLE DATABASE                NO  NO  NO

PDB_DBA
CREATE SESSION                           NO  NO  NO


SQL>
```

mengambil dan menampilkan daftar semua System Privilege yang ada di dalam Role bernama PDB_DBA

SQL> select * from DBA_role_PRIVS

where GRANTEE='PDB_DBA';

```
SQL> select * from DBA_role_PRIVS
where GRANTEE='PDB_DBA';
  2
GRANTEE
--------------------------------------------------------------------------------
GRANTED_ROLE
--------------------------------------------------------------------------------
ADM DEL DEF COM INH
--- --- --- --- ---
PDB_DBA
CONNECT
NO  NO  YES NO  NO

PDB_DBA
WM_ADMIN_ROLE
YES NO  YES YES YES

GRANTEE
--------------------------------------------------------------------------------
GRANTED_ROLE
--------------------------------------------------------------------------------
ADM DEL DEF COM INH
--- --- --- --- ---


SQL>
```

Mengambil dan menampilkan daftar Role yang diberikan ke Role bernama PDB_DBA

SQL> SELECT * FROM ROLE_SYS_PRIVS

WHERE ROLE='CONNECT';

```
SQL> SELECT * FROM ROLE_SYS_PRIVS
WHERE ROLE='CONNECT';
  2
ROLE
--------------------------------------------------------------------------------
PRIVILEGE                                ADM COM INH
---------------------------------------- --- --- ---
CONNECT
SET CONTAINER                            NO  YES YES

CONNECT
CREATE SESSION                           NO  YES YES


SQL>
```

Mengambil dan menampilkan semua System Privilege yang ada di dalam Role bernama CONNECT

==========

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> select * from dba_PROFILES

where PROFILE='DEFAULT';

```
SQL> select * from dba_PROFILES
where PROFILE='DEFAULT';
  2
PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
FAILED_LOGIN_ATTEMPTS            PASSWORD
10
NO  NO  NO  YES NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
COMPOSITE_LIMIT                  KERNEL
```

```
------------------------------ --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
INACTIVE_ACCOUNT_TIME           PASSWORD
365
NO  NO  NO  YES NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                   RESOURCE
------------------------------ --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
PASSWORD_ROLLOVER_TIME          PASSWORD
0
NO  NO  NO  YES NO  NO


18 rows selected.

SQL>
```

Mengambil dan menampilkan semua pengaturan sumber daya dan kata sandi (password) yang ditetapkan pada Profile bernama DEFAULT

SQL> create profile C##GENERAL

limit

SESSIONS_PER_USER 4

IDLE_TIME 15

FAILED_LOGIN_ATTEMPTS 3

PASSWORD_LIFE_TIME 180

container=all;

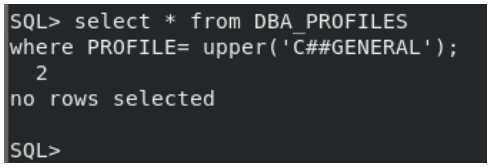
```
SQL> create profile C##GENERAL
limit
SESSIONS_PER_USER 4
IDLE_TIME 15
FAILED_LOGIN_ATTEMPTS 3
PASSWORD_LIFE_TIME 180
container=all;
  2    3    4    5    6    7  create profile C##GENERAL
*
ERROR at line 1:
ORA-65040: Operation is not allowed from within a pluggable database.
Help: https://docs.oracle.com/error-help/db/ora-65040/


SQL>
```

membuat Common Profile (C##GENERAL) yang berlaku di semua PDB (container=all), menetapkan batasan keamanan seperti 4 sesi per user, 15 menit idle time, 3 kali percobaan login gagal, dan masa berlaku password 180 hari. Error karena ORA-65040: Operation is not allowed from within a pluggable database. Error karena DDL yang membuat Common (yang berlaku di seluruh container dengan CONTAINER=ALL) hanya boleh dijalankan ketika terhubung ke CDB$ROOT (root container), bukan dari Pluggable Database (PDB) yang sekarang sedang dibuka

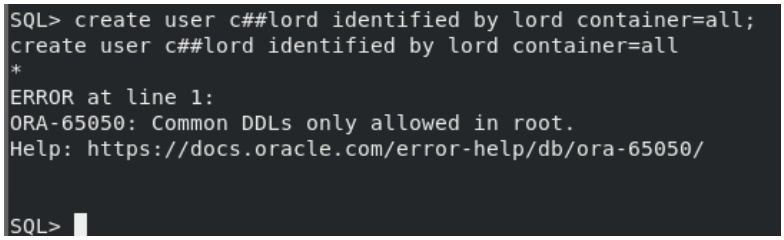
SQL> select * from DBA_PROFILES

where PROFILE= upper('C##GENERAL');



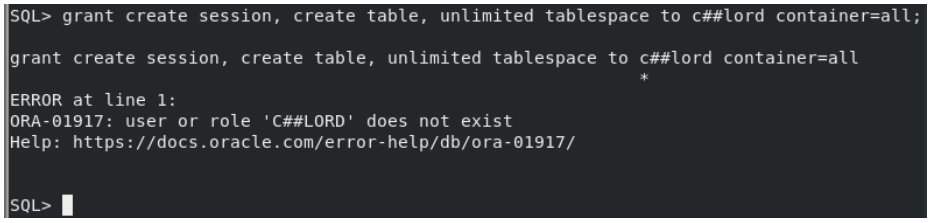
Mengambil dan menampilkan semua pengaturan kebijakan keamanan (resource dan password) yang ada di Common Profile bernama C##GENERAL. Fungsi UPPER() untuk memastikan nama profile dicocokkan dengan benar. Hasilnya no rows selected, karena profile bernama C##GENERAL tidak ditemukan di container database saat ini, karena saat membuat Common Profile sebelumnya error

SQL> create user c##lord identified by lord container=all;



membuat Common User (C##LORD) dengan password lord yang dapat terhubung ke semua PDB (container=all) di CDB. Gagal karena ORA-65050: Common DDLs only allowed in root. Error karena DDL yang membuat objek Common (yaitu, yang berlaku di seluruh container melalui CONTAINER=ALL) hanya dapat dijalankan saat terhubung ke CDB$ROOT (root container), bukan dari Pluggable Database (PDB) tempat saat ini terhubung

SQL> grant create session, create table, unlimited tablespace to c##lord container=all;



memberikan System Privilege (CREATE SESSION, CREATE TABLE) dan hak penyimpanan (UNLIMITED TABLESPACE) ke Common User C##LORD, dan menetapkan hak tersebut berlaku di semua container (container=all). Gagal karena ORA-01917: user or role 'C##LORD' does not exist. Error karena user C##LORD tidak ditemukan. Cara sebelumnya untuk membuat user ini gagal (DDL Common harus dilakukan dari CDB$ROOT), sehingga user tidak pernah terdaftar

SQL> select * from DBA_users

where USERNAME= upper('c##lord');

```
SQL> select * from DBA_users
where USERNAME= upper('c##lord');
  2
no rows selected

SQL>
```

mengambil semua detail dari view DBA_USERS untuk user C##LORD. Fungsi UPPER() memastikan nama user dicocokkan dengan benar. Hasilnya no rows selected. Ini menunjukkan bahwa user bernama C##LORD tidak terdaftar dalam container database saat ini. Sebelumnya, cara untuk membuat Common User (C##LORD) error karena melanggar aturan DDL (DDL Common harus dijalankan dari CDB$ROOT), sehingga user tidak pernah terdaftar

SQL> alter user c##lord profile C##GENERAL container=all;

```
SQL> alter user c##lord profile C##GENERAL container=all;
alter user c##lord profile C##GENERAL container=all
           *
ERROR at line 1:
ORA-01918: user 'C##LORD' does not exist
Help: https://docs.oracle.com/error-help/db/ora-01918/


SQL>
```

memindahkan Common Profile bernama C##GENERAL ke Common User C##LORD, dan menetapkan perubahan ini berlaku di semua container (container=all). Meskipun perintah ini benar, sebelumnya (dimana user C##LORD dan profile C##GENERAL gagal dibuat karena DDL Common dijalankan dari PDB), perintah ini akan gagal karena ORA-01917: user or role/profile does not exist. Fungsi sebenarnya: Jika berhasil, perintah ini akan memaksa user C##LORD untuk mematuhi kebijakan keamanan yang ditetapkan dalam profile C##GENERAL (seperti batas sesi dan masa berlaku password) di setiap PDB yang ada

SQL> select * from DBA_users

where USERNAME= upper('c##lord');

```
SQL> select * from DBA_users
where USERNAME= upper('c##lord');
  2
no rows selected

SQL>
```

mengambil semua *view* **DBA_USERS** untuk *user* bernama **C##LORD**. Fungsi UPPER() memastikan nama *user* dicocokkan dengan benar. Hasilnya **no rows selected**. Ini memastikan *user* bernama **C##LORD tidak terdaftar** dalam *container* database sekarang. Sesuai dengan *error* sebelumnya, di mana cara untuk membuat *Common User* (C##LORD) gagal karena melanggar aturan DDL (*Common* DDL harus dijalankan dari **CDB$ROOT**), sehingga *user* tidak pernah terdaftar

==========

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> alter session set container=freepdb1;

```
SQL> alter session set container=freepdb1;

Session altered.

SQL>
```

Berpindah ke PDB bernama FREEPDB1

SQL> show con_name;

```
SQL> show con_name;

CON_NAME
------------------------------
FREEPDB1
SQL>
```

Menampilkan nama container saat ini: FREEPDB1

SQL> select * from dba_PROFILES

where PROFILE='DEFAULT';

```
SQL> select * from dba_PROFILES
where PROFILE='DEFAULT';
  2
PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
FAILED_LOGIN_ATTEMPTS            PASSWORD
10
NO  NO  NO  YES NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
COMPOSITE_LIMIT                  KERNEL
```

```
------------------------------ --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
INACTIVE_ACCOUNT_TIME            PASSWORD
365
NO  NO  NO  YES NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
------------------------------ --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
DEFAULT
PASSWORD_ROLLOVER_TIME           PASSWORD
0
NO  NO  NO  YES NO  NO


18 rows selected.

SQL>
```

Mengambil dan menampilkan semua pengaturan kebijakan keamanan (resource dan password) yang ditetapkan di Profile bernama DEFAULT

SQL> create profile freepdb1profile

limit

SESSIONS_PER_USER UNLIMITED

IDLE_TIME UNLIMITED

FAILED_LOGIN_ATTEMPTS 10

PASSWORD_LIFE_TIME 180

INACTIVE_ACCOUNT_TIME 10;

```
SQL> create profile freepdb1profile
limit
SESSIONS_PER_USER UNLIMITED
IDLE_TIME UNLIMITED
FAILED_LOGIN_ATTEMPTS 10
PASSWORD_LIFE_TIME 180
INACTIVE_ACCOUNT_TIME 10;
  2    3    4    5    6    7  create profile freepdb1profile
*
ERROR at line 1:
ORA-02377: invalid profile limit INACTIVE_ACCOUNT_TIME
Help: https://docs.oracle.com/error-help/db/ora-02377/


SQL>
```

membuat Local Profile bernama FREEPDB1PROFILE dengan beberapa batasan keamanan (limit), seperti tidak terbatasnya sesi aktif dan idle time, 10 kali percobaan login gagal, masa berlaku password 180 hari, dan batas 10 hari sebelum akun dianggap tidak aktif. Gagal karena ORA-02377: invalid profile limit INACTIVE_ACCOUNT_TIME. Ini error karena parameter INACTIVE_ACCOUNT_TIME yang digunakan dalam LIMIT tidak valid saat ini sebagai batas profil yang berdiri sendiri

SQL> create profile freepdb1profile

limit

SESSIONS_PER_USER UNLIMITED

IDLE_TIME UNLIMITED

FAILED_LOGIN_ATTEMPTS 10

PASSWORD_LIFE_TIME 180

INACTIVE_ACCOUNT_TIME 15;

```
SQL> create profile freepdb1profile
limit
SESSIONS_PER_USER UNLIMITED
IDLE_TIME UNLIMITED
FAILED_LOGIN_ATTEMPTS 10
PASSWORD_LIFE_TIME 180
INACTIVE_ACCOUNT_TIME 15;
  2    3    4    5    6    7
Profile created.

SQL>
```

Membuat Local Profile bernama FREEPDB1PROFILE dengan beberapa batasan keamanan (limit)

SQL> select * from DBA_PROFILES

where PROFILE= upper('freepdb1profile');

```
SQL> select * from DBA_PROFILES
where PROFILE= upper('freepdb1profile');
  2
PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
FREEPDB1PROFILE
FAILED_LOGIN_ATTEMPTS            PASSWORD
10
NO  NO  NO  NO  NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
FREEPDB1PROFILE
COMPOSITE_LIMIT                  KERNEL
```

```
------------------------------ --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
FREEPDB1PROFILE
INACTIVE_ACCOUNT_TIME           PASSWORD
15
NO  NO  NO  NO  NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                   RESOURCE
------------------------------ --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
FREEPDB1PROFILE
PASSWORD_ROLLOVER_TIME          PASSWORD
DEFAULT
NO  NO  NO  NO  NO  NO


18 rows selected.

SQL>
```

Mengambil dan menampilkan semua pengaturan kebijakan keamanan (resource dan password) yang ditetapkan dalam Local Profile bernama FREEPDB1PROFILE

SQL> create user sara identified by sara profile freepdb1profile;

```
SQL> create user sara identified by sara profile freepdb1profile;

User created.

SQL>
```

Membuat User Lokal baru bernama SARA dan mengaitkannya dengan Local Profile FREEPDB1PROFILE

SQL> grant create session, create table, unlimited tablespace to sara;

```
SQL> grant create session, create table, unlimited tablespace to sara;

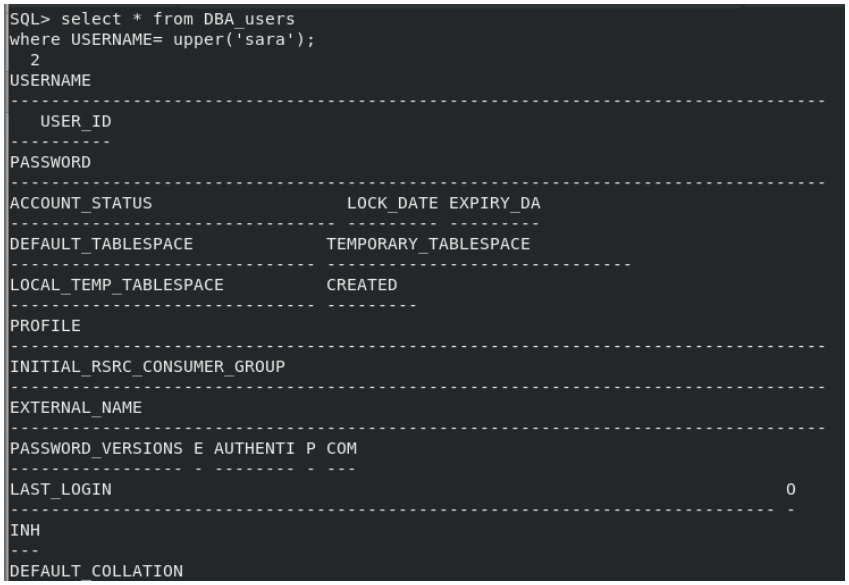
Grant succeeded.

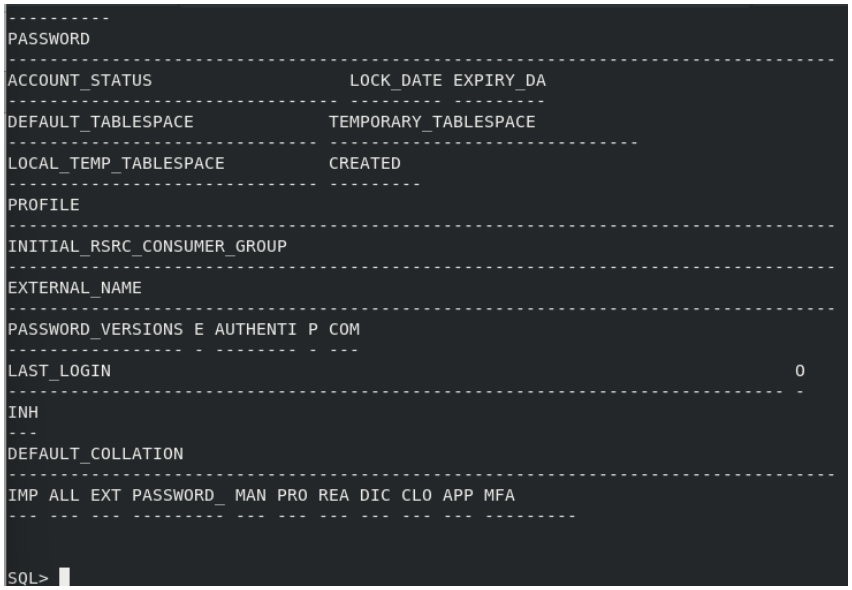
SQL>
```

Memberikan hak CREATE SESSION, CREATE TABLE, dan UNLIMITED TABLESPACE ke User Lokal SARA

SQL> select * from DBA_users

where USERNAME= upper('sara');

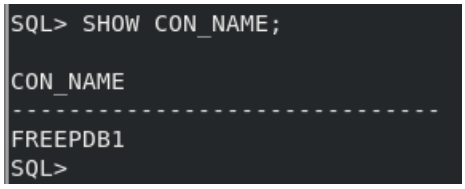




Mengambil dan menampilkan semua detail dari view DBA_USERS untuk user bernama SARA

==========

SQL> SHOW CON_NAME;

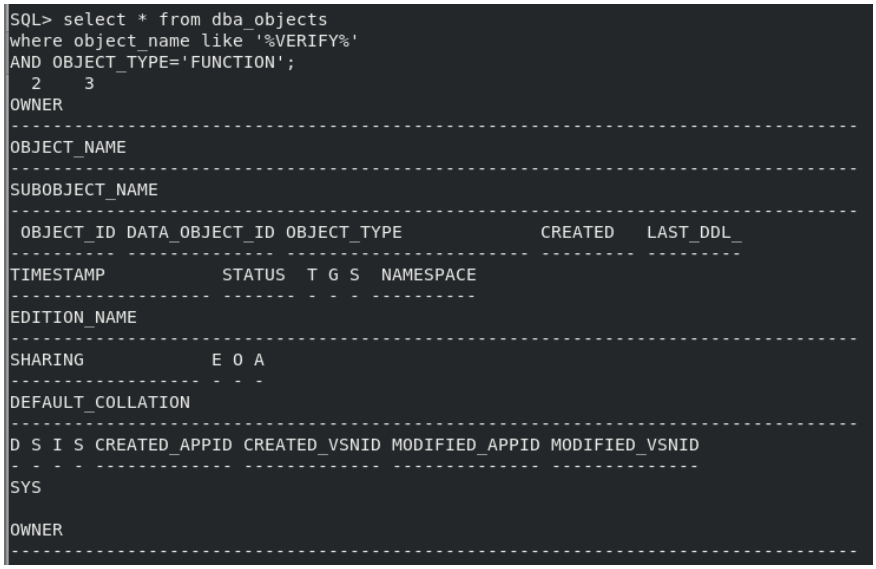


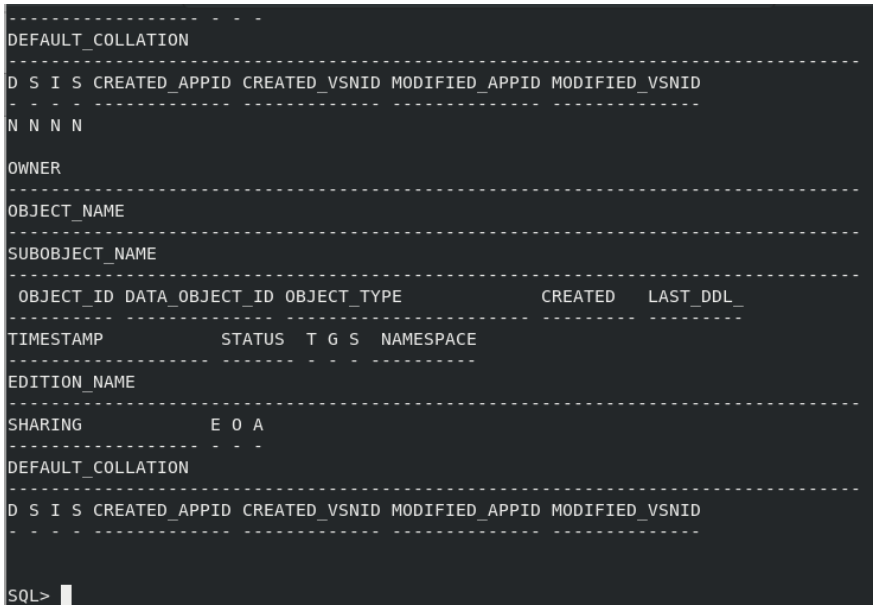
Menampilkan nama container saat ini: FREEPDB1

SQL> select * from dba_objects

where object_name like '%VERIFY%'

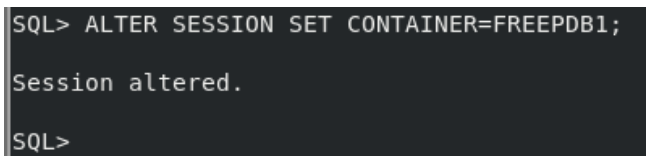
AND OBJECT_TYPE='FUNCTION';





Mengambil dan menampilkan semua function yang namanya mengandung string VERIFY

SQL> ALTER SESSION SET CONTAINER=FREEPDB1;

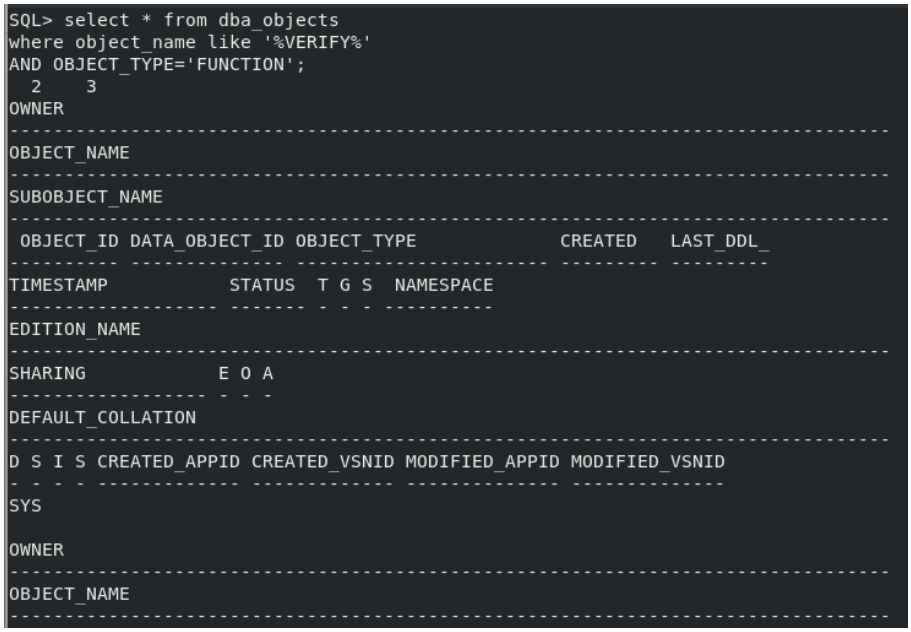


Berpindah ke PDB bernama FREEPDB1

SQL> select * from dba_objects

where object_name like '%VERIFY%'

AND OBJECT_TYPE='FUNCTION';

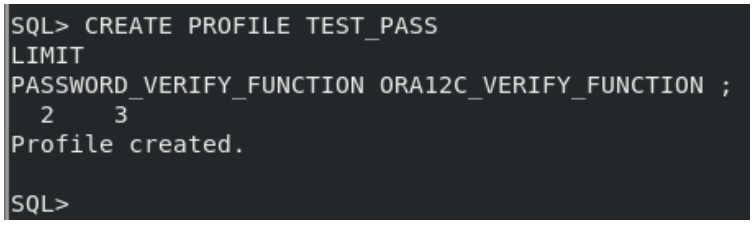


Mengambil dan menampilkan semua function yang namanya mengandung string VERIFY

SQL> CREATE PROFILE TEST_PASS

LIMIT

PASSWORD_VERIFY_FUNCTION ORA12C_VERIFY_FUNCTION ;



Membuat Profile bernama TEST_PASS dan mewajibkan semua user yang menggunakan profile ini untuk mematuhi aturan yang ditetapkan oleh Fungsi Verifikasi Password yaitu ORA12C_VERIFY_FUNCTION

SQL> SELECT * FROM DBA_PROFILES

WHERE PROFILE='TEST_PASS';

```
SQL> SELECT * FROM DBA_PROFILES
WHERE PROFILE='TEST_PASS';
  2
PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
TEST_PASS
PASSWORD_VERIFY_FUNCTION         PASSWORD
ORA12C_VERIFY_FUNCTION
NO  NO  NO  NO  NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
TEST_PASS
COMPOSITE_LIMIT                  KERNEL
```

```
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
TEST_PASS
INACTIVE_ACCOUNT_TIME            PASSWORD
DEFAULT
NO  NO  NO  NO  NO  NO


PROFILE
--------------------------------------------------------------------------------
RESOURCE_NAME                    RESOURCE
-------------------------------- --------
LIMIT
--------------------------------------------------------------------------------
COM INH IMP ORA MAN APP
--- --- --- --- --- ---
TEST_PASS
PASSWORD_ROLLOVER_TIME           PASSWORD
DEFAULT
NO  NO  NO  NO  NO  NO


18 rows selected.

SQL>
```

Mengambil dan menampilkan semua pengaturan kebijakan keamanan (resource dan password) yang ditetapkan dalam Profile bernama TEST_PASS

SQL> create user test100 identified by welcome profile TEST_PASS;

```
SQL> create user test100 identified by welcome  profile TEST_PASS;
create user test100 identified by welcome  profile TEST_PASS
*
ERROR at line 1:
ORA-28003: The password chosen did not meet the required complexity rules set
by your organization.
ORA-20000: password length less than 8 characters
Help: https://docs.oracle.com/error-help/db/ora-28003/


SQL>
```

membuat User Lokal TEST100 dengan password welcome, dan menetapkan bahwa user tersebut harus mengikuti kebijakan keamanan profile TEST_PASS. Error karena ORA-28003: The password chosen did not meet the required complexity rules set by your organization, ORA-20000: password length less than 8 characters. Error karena password yang dipilih (welcome) tidak memenuhi aturan yang diwajibkan oleh TEST_PASS (yang menggunakan fungsi verifikasi password ORA12C_VERIFY_FUNCTION). Aturan yang dilanggar adalah panjang password kurang dari 8 karakter

SQL> create user test100 identified by welcome1234 profile TEST_PASS;

```
SQL> create user test100 identified by welcome1234  profile TEST_PASS;
create user test100 identified by welcome1234  profile TEST_PASS
*
ERROR at line 1:
ORA-28003: The password chosen did not meet the required complexity rules set
by your organization.
ORA-20000: password must contain 1 or more special characters
Help: https://docs.oracle.com/error-help/db/ora-28003/


SQL>
```

membuat User Lokal TEST100 dengan password welcome1234, dan menetapkan bahwa user harus mengikuti kebijakan keamanan profile TEST_PASS. Error karena ORA-28003: The password chosen did not meet the required complexity rules set by your organization, ORA-20000: password must contain 1 or more special characters. Error karena password yang dipilih (welcome1234) tidak memenuhi aturan yang diwajibkan oleh TEST_PASS (yang menggunakan fungsi verifikasi password ORA12C_VERIFY_FUNCTION). Aturan yang dilanggar adalah kewajiban menyertakan setidaknya 1 karakter khusus

SQL> create user test100 identified by welcome1234# profile TEST_PASS;

```
SQL> create user test100 identified by welcome1234#  profile TEST_PASS;

User created.

SQL>
```

membuat User Lokal TEST100 dengan password welcome1234#, dan menetapkan user harus mengikuti kebijakan keamanan profile TEST_PASS

SQL> grant create session to test100;

```
SQL> grant create session to test100;

Grant succeeded.

SQL>
```

Memberikan System Privilege CREATE SESSION ke user TEST100